

Nathan D. Hart

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EDUCATION:

University of Maryland

B.Eng in Mechanical Engineering

College Park Scholars: Data Justice

GPA: 3.917

College Park, MD

Aug. 2025-May 2029

Aug. 2025-May 2027

Technical Experience:

University of Maryland Engineering Design Project

Project Sub-Team Leader

College Park, MD

Sep.-Dec. 2025

- Led a 3 person sub-team in addressing mission specific design constraints and goals
- Developed and presented 3 comprehensive design presentations and two 30+ page design reports
- Utilized Onshape CAD software to design and 3D print effective vehicle components
- Collaborated with 3 distinct sub-teams to accomplish objectives

Wind Terpiners Competition team

Drivetrain Team Leader

College Park, MD

Sep. 2025-Present

- Worked alongside 4 engineers to establish a fundamental approach for drivetrain of the team's wind turbine
- Researched and conceptualized turbine components using Solidworks 3D Design application, FEA analysis
- Developed and iterated on initial design using various manufacturing methods (3D printing, metalworking)
- Coordinated with 2 other sub-teams; achieved 2nd place in technical design and testing at eastern regional

CAD Assembly Project

Project Team Member

Chelmsford, MA

May 2024-Jun. 2025

- Developed comprehensive car jack assembly with 35+ unique components and sub assemblies
- Used FEA to evaluate component effectiveness; showcased system abilities through solidworks animation
- Experimented with Autodesk Fusion generative design to iterate components and optimize design
- Collaborated with 5 team members to showcase work in a final project presentation

WORK EXPERIENCE:

Middlesex Sheriff's Office

Youth Public Service Academy Counselor

Chelmsford, MA

Jun.-Aug. 2024, 2025

- Mentored over 80 children from 20+ communities across Massachusetts
- Partnered with 30+ other counselors to maintain a safe and productive environment for 5 camp sessions
- Organized and monitored general activities, mealtime food distribution 2x daily, and weekly vehicle transport of camp equipment

US Officials / CYSA Refereeing

Head Official / Assistant Referee

Chelmsford, MA, State of New Hampshire

Sep. 2020-Nov. 2024

- Refereed over 250+ matches for age groups ranging from 7 to 19 years old; Obtained valuable communication and personal skills by managing the behavior of coaches, players, and fans
- Officiated at proficiency levels ranging from youth club/recreational to professional academy teams

Additional Information:

Honors and Awards:

University of Maryland President's Scholarship

Awarded Feb. 2025

Skills and Certifications:

Proficient: Solidworks, Autodesk Fusion, Onshape, Microsoft Office, Google Workspace